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Understanding UX in Digital Transformation

A Service Design Approach in Traditional Manufacturing

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Abstract

As traditional manufacturing firms adopt digital systems to modernize operations, misalignments often arise between how these systems are designed and how work actually happens on the shop floor. This thesis examines how such service-related factors shape user experience (UX) and influence the adoption of digital technologies among operators, engineers, and managers in a mid-sized furniture factory in China. Using a qualitative single-case study grounded in service design principles and Reflexive Thematic Analysis (RTA), the research draws on interviews, field observations, and co-creation workshops to explore how users navigate, adapt to, and engage with digital tools in their daily routines.

The study identifies four interconnected service challenges: misaligned workflows, insufficient feedback mechanisms, cognitive overload, and misunderstandings across roles. These issues reveal that digital adoption is not just a matter of usability or training, but also depends on emotional experience, trust, and organizational communication. By treating internal systems as employee-facing services, the thesis expands the application of service design to industrial UX, highlighting the situated and socially embedded nature of digital transformation.

On a practical level, the findings offer actionable insights—such as embedding feedback and error recovery into interfaces, aligning digital processes with real production tasks, and establishing regular cross-role dialogue. Ultimately, the study suggests that sustainable digital transformation requires more than technological capability: it calls for a rethinking of how people, systems, and organizations evolve together through shared understanding and participatory design.

Keywords: User Experience (UX), Service Design, Digital Transformation, Manufacturing

Sammanfattning

När traditionella tillverkningsföretag inför digitala system för att modernisera sina verksamheter, uppstår ofta en bristande överensstämmelse mellan hur systemen är designade och hur arbetet faktiskt utförs på verkstadsgolvet. Denna avhandling undersöker hur sådana tjänsterelaterade faktorer påverkar användarupplevelsen (UX) och inverkar på införandet av digital teknik bland operatörer, ingenjörer och chefer i en medelstor möbelfabrik i Kina. Studien bygger på en kvalitativ enfallsstudie, grundad i servicedesignprinciper och reflexiv tematisk analys (RTA), och baseras på intervjuer, fältobservationer och samskapande workshops för att förstå hur användare navigerar, anpassar sig till och arbetar med digitala verktyg i sin dagliga praktik.

Studien identifierar fyra sammanlänkade utmaningar relaterade till tjänstedesign: felanpassade arbetsflöden, brist på återkopplingsmekanismer, kognitiv överbelastning och rollbaserade missförstånd. Dessa visar att införandet av digitala system inte bara handlar om användbarhet eller utbildning, utan också om känslomässiga upplevelser, förtroende och intern kommunikation. Genom att behandla interna system som tjänster riktade mot anställda, utvidgar avhandlingen servicedesignens tillämpning till industriell UX och betonar den kontextuella och socialt inbäddade karaktären av digital transformation.

På en praktisk nivå erbjuder resultaten konkreta designinsikter—såsom att integrera återkoppling och felhantering i gränssnitt, anpassa digitala processer till verkliga arbetsflöden och etablera kontinuerlig dialog mellan olika roller. Sammanfattningsvis visar studien att hållbar digital transformation kräver mer än teknisk kapacitet: det handlar om att omforma hur människor, system och organisationer utvecklas tillsammans genom delad förståelse och deltagande design.

Nyckelords: Användarupplevelse (UX), Tjänstedesign, Digital transformation, Tillverkningsindustri

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Contents

1 Introduction **1**
- 1.1 Background . . . 2
- 1.2 Problem Formulation . . . 2
- 1.3 Research Goal and Research Question . . . 3
- 1.4 Research Contributions . . . 4
- 1.5 Methodological Overview . . . 4
- 1.6 Delimitations . . . 5
- 1.7 Structure of the Thesis . . . 5

2 Literature Review **6**
- 2.1 Theoretical Foundations and Human-Centered Approaches . . . 6
  - 2.1.1 Service Design as a Human-Centered Approach . . . 6
  - 2.1.2 Digital Transformation and UX Challenges in Manufacturing . 7
  - 2.1.3 Human and Organizational Dynamics in Adoption . . . 7
- 2.2 Highly Relevant Work . . . 8
  - 2.2.1 Service Design and Human-Centered Approaches in Digital Transformation . . . 8
  - 2.2.2 UX Challenges and Adoption on the Shop Floor . . . 8
  - 2.2.3 Cross-Role Communication and Cultural Factors in Adoption 9

3 Research Design and Philosophical Approach **10**
- 3.1 Research Design and Theoretical Framework . . . 10
- 3.2 Qualitative Research Rationale . . . 10
- 3.3 Case Study Approach . . . 10
- 3.4 Philosophical Paradigm: Interpretivism . . . 11
- 3.5 Reflexivity and Positionality . . . 11
- 3.6 Case Context and Stakeholder Landscape . . . 11
  - 3.6.1 Digital Transformation in the Factory . . . 11
  - 3.6.2 Organizational Structure and Stakeholder Roles . . . 12
  - 3.6.3 Rationale for Case Selection . . . 13
- 3.7 Data Collection Procedures . . . 13
- 3.8 Data Analysis and Trustworthiness . . . 14
- 3.9 Ethical Considerations and Data Governance . . . 14

4 Findings **15**
- 4.1 Overview of the Analytical Themes . . . 15
- 4.2 Theme 1: System–Workflow Misalignment . . . 16
- 4.3 Theme 2: Feedback Gaps and Error Anxiety . . . 17
- 4.4 Theme 3: Learning Curves and Cognitive Load . . . 18

4.5 Theme 4: Cross-Role Misunderstandings 20
4.5.1 Operators: Task-Focused and Risk-Aware 20
4.5.2 Engineers and Managers: System-Centered and Outcome-Focused 20
4.5.3 Workshop Insights: Surfacing the Disconnects 21
4.6 Summary and Thematic Interrelations 22

5 Discussion **24**
5.1 Theoretical Integration of Findings 24
5.1.1 Emotional Responsiveness and Cognitive Support 24
5.1.2 Workflow Alignment and Service Continuity 24
5.1.3 Cross-Role Coordination and Shared Understanding 25
5.1.4 Trust Infrastructure and Perceived Safety 25
5.2 Practical and Design Implications 26
5.3 Methodological Reflection 27
5.4 Future Research Directions 28

6 Conclusion **29**

References **29**

A (Interview Question Guid) **35**
Appendix A: Interview Question Guide 35

List of Figures

4.1 Service Blueprint: Misalignment Between Workflow and Digital System 17
4.2 Journey map: operator experience with digital system interaction. . . 19

List of Tables

3.1 Overview of Interview Participants 13

4.1 Overview of Analytical Themes 16
4.2 Thematic summary: emerging UX patterns and stakeholder impacts . 22

List of Acronyms and Abbreviations

AGV	Automated Guided Vehicle system
ERP	Enterprise Resource Planning
HCI	Human-Computer Interaction
RTA	Reflexive Thematic Analysis
SME	Small and Medium-sized Enterprise
UX	User Experience

1

Introduction

Digital transformation has become a crucial driver of change across industries, reshaping the way organizations operate, compete, and deliver value[12]. In the manufacturing sector, particularly within small- and medium-sized enterprises (SMEs), the push toward digitalization presents both unprecedented opportunities and complex challenges. While advanced technologies such as automation, real-time data analytics, and cloud-based systems promise significant productivity gains, their integration into traditional manufacturing settings often encounters barriers beyond technical implementation alone[9, 18].

Rather than being purely a technological endeavor, successful digital transformation hinges on human and organizational factors, including user acceptance, workflow alignment, and cross-functional collaboration[2, 31]. In this context, the role of user experience (UX) design has emerged as pivotal: poorly designed digital systems can lead to low adoption rates, operational inefficiencies, and employee resistance, ultimately undermining the intended benefits of technological investments[26].

Service design, with its human-centered and holistic approach, offers a promising lens for addressing these challenges[27]. By focusing on the needs, experiences, and behaviors of frontline users, service design not only enhances system usability but also aligns digital tools with real-world workflows and organizational culture. Applying service design principles in manufacturing environments—especially in SMEs—provides a critical opportunity to bridge the gap between technology and people, fostering more sustainable and inclusive digital transformations[16].

This thesis investigates how service-related factors influence the user experience and adoption of digital systems in a mid-sized furniture factory in China. Building on interdisciplinary insights from service design, UX research, and digital transformation studies, the research aims to develop a deeper understanding of the interplay between human, organizational, and technological dimensions in manufacturing digitalization efforts. By capturing the lived experiences of operators, engineers, and managers, the study seeks to generate practical recommendations for designing digital systems that are not only technically effective but also socially and operationally sustainable.

The following sections provide the necessary context, including a review of relevant literature (Chapter 2), a detailed explanation of the research methodology (Chapter 3), an empirical analysis of the findings (Chapter 4), a discussion connecting results to broader theoretical and practical implications (Chapter 5), and concluding reflections with suggestions for future research (Chapter 6).

1.1 Background

The manufacturing sector is indeed experiencing significant changes due to the integration of Industry 4.0 technologies, which encompass data-driven interfaces, automation, and cloud-based systems. The literature indicates that Industry 4.0 represents a suite of enabling technologies that facilitate the transformation of industrial processes into smart ecosystems where humans, machines, and devices interact efficiently to enhance productivity and profitability[23]. However, integrating these technologies into traditional manufacturing environments—often characterized by legacy systems and established workflows—presents significant human and organizational challenges. Research highlights that factors such as resistance to change, skills gaps, and misaligned priorities frequently impede successful digital transformation [1].

Mid-sized manufacturing enterprises, such as furniture factories, are increasingly adopting digital tools to optimize production processes. Examples include touchscreen workstations for operator input, real-time monitoring dashboards for production oversight, and barcode-based tracking systems for inventory management. While these systems are designed to streamline operations, their implementation often fails to account for the practical realities faced by frontline workers. Operators and shop floor personnel, who interact with these systems daily, may find them complex, unintuitive, or disconnected from their existing workflows. The misalignment between digital transformation initiatives and employee acceptance can indeed lead to various negative outcomes, including low adoption rates and user resistance. Research indicates that the success of digital transformation is significantly influenced by employees' mindsets and their willingness to embrace new technologies. If employees do not possess an appropriate mindset for digital transformation, it can result in operational errors and increased cognitive load, ultimately undermining the intended benefits of these initiatives[21].

To address these challenges, service design has emerged as a human-centric methodology that aligns technical systems with user needs. Unlike traditional design approaches that focus solely on interfaces or functionality, service design takes a holistic view, considering the entire ecosystem of people, processes, and touchpoints involved in a system's use. Key principles include co-creation, stakeholder alignment, and the orchestration of seamless interactions across touchpoints[44]. In manufacturing, service design can simplify interfaces, provide just-in-time training, align digital tools with workflows, and foster cross-role dialogue, thereby improving adoption and operational outcomes[5, 38].

This thesis investigates how service-related factors influence the user experience (UX) and adoption of digital systems in a medium-sized furniture factory in China, analyzed through a service design lens. By focusing on the experiences of frontline workers, the study aims to bridge the gap between technological capabilities and human needs, offering insights into creating more resilient and humane workplaces during digital transformation.

1.2 Problem Formulation

Despite significant investments in digital tools, traditional manufacturing companies frequently face substantial barriers to effective adoption[40]. Frontline opera-

tors often find these systems unintuitive, resulting in resistance and inefficient use. Engineers may struggle to maintain tools developed without sufficient user input, while managers frequently lack the insights needed to address the underlying causes of shop floor resistance. These challenges are not solely technical but are systemic, rooted in service-related issues such as organizational culture, communication breakdowns, and inadequate user experience design. For example, research indicates that frontline workers often encounter a "technology gap", where digital tools fail to align with their practical needs, leading to reduced productivity and engagement[19]. The absence of worker involvement in the design process further exacerbates these issues, as tools are often prioritized for technical functionality over usability[3].

Current research on digital transformation in manufacturing largely centers on technology deployment, business model innovation, and strategies at the enterprise scale [11]. Yet, the lived experiences of frontline workers—particularly within small- and medium-sized enterprises (SMEs)—remain insufficiently explored, despite their central role in production and frequent exposure to resource limitations. Studies emphasize that employee engagement and organizational support significantly impact job satisfaction and performance, especially in flexible manufacturing environments [10]. When workers feel supported and part of the organization, their motivation and productivity tend to rise.Job crafting—the ability for employees to shape aspects of their roles—has also been linked to stronger engagement, innovation, and career satisfaction. Encouraging such autonomy not only boosts individual fulfillment but also strengthens the alignment between digital initiatives and employee needs, underscoring the importance of human-centered management [35].While service design, with its focus on co-creation, stakeholder alignment, and integrated user experiences, presents a compelling framework for inclusive digital solutions, its application within manufacturing SMEs remains underdeveloped. Existing studies in digital servitization suggest that shifting from product-based to service-oriented models requires careful coordination of sociotechnical resources and active value co-creation [13]. However, research addressing how service design can enhance internal systems and employee-facing technologies is still limited, marking a clear gap in current literature.

This thesis seeks to address these gaps by examining how service-related factors influence the user experience (UX) and adoption of digital systems in a medium-sized furniture factory in China. Utilizing a service design approach, the study aims to identify specific barriers faced by frontline workers, engineers, and managers, and to propose human-centered solutions that enhance digital transformation in traditional manufacturing environments.

1.3 Research Goal and Research Question

The primary goal of this research is to examine how service-related factors shape the user experience (UX) and influence the adoption of digital systems among operators, engineers, and managers in a traditional manufacturing environment. By adopting a human-centered perspective, the study seeks to identify barriers and enablers of digital transformation, contributing to the development of more inclusive and effective digital strategies for SMEs. This goal underscores the importance of aligning technological advancements with the practical needs and experiences of shop floor stakeholders.

Research Question: What service-related factors influence the user experience and adoption of digital systems in traditional manufacturing environments?

1.4 Research Contributions

This thesis contributes to both theoretical development and practical implementation in the intersecting fields of service design, UX, and industrial digital transformation. On a theoretical level, the research advances the application of service design by situating its principles—such as co-creation, stakeholder alignment, and touchpoint orchestration—within the context of employee-facing systems in traditional manufacturing. Whereas service design has historically focused on customer-facing experiences, this study demonstrates its relevance for internal users navigating complex production workflows. By framing internal digital tools as services that unfold across human interactions and decision-making layers, the research brings new nuance to how value is co-produced in digitally evolving organizations. In doing so, it complements and deepens emerging discussions on digital servitization, adding empirical texture to how service design can support collaboration, contextual usability, and bottom-up innovation in SMEs.

Practically, the study offers a set of grounded insights for improving system usability and digital integration in industrial environments. These insights are not presented as prescriptive checklists, but as interpretive frameworks derived from the lived experiences of operators, engineers, and managers within a real production context. Through detailed engagement with field data—including interviews, observations, and a co-creation workshop—the research surfaces how digital tools are often appropriated, avoided, or reconfigured through informal workarounds and social negotiation. These insights inform a human-centered UX improvement approach, one that places emphasis on intuitive system design, adaptive support structures, and iterative feedback mechanisms. Rather than proposing a one-size-fits-all model, the contribution lies in showing how digital systems must be embedded within broader service ecosystems that accommodate ambiguity, tacit knowledge, and ongoing organizational learning. This perspective is particularly valuable for SMEs where digital transformation initiatives are often constrained by limited resources and low design maturity, yet still depend on the trust, skill, and adaptability of human users.

1.5 Methodological Overview

This research is structured as a qualitative single-case study conducted in a mid-sized furniture factory in Anhui, China, with approximately 180 employees. The single-case study design enables an in-depth, context-sensitive exploration of the interplay between service design, UX, and digital transformation, which is particularly suited for understanding complex, human-centered challenges in a specific organizational setting. The study employs a service design lens to investigate system usability and workflow integration, utilizing a multi-method data collection approach:

- Semi-structured interviews with 14 participants (7 operators, 3 engineers, 4 managers) to capture diverse stakeholder perspectives on digital system use.

- Non-participant observations across five production stages to document real-world workflows and system interactions.
- Co-creation workshops with seven participants, employing service design tools such as journey maps and service blueprints to collaboratively identify UX challenges and ideate solutions.
- Reflexive Thematic Analysis (RTA)[8] to interpret the data, allowing for a systematic identification of themes while incorporating researcher reflexivity to address potential biases.

This participatory and multi-faceted methodology ensures a comprehensive understanding of UX challenges from multiple stakeholder viewpoints, aligning with the study's human-centered focus. The approach is grounded in established qualitative research practices, which prioritize depth and stakeholder engagement (Qualitative Research Methods).

1.6 Delimitations

The study is limited to a single mid-sized furniture factory in China, focusing exclusively on the adoption of internal digital systems, such as touchscreen workstations and real-time monitoring dashboards, used by operators, engineers, and managers. It does not cover customer-facing technologies, such as e-commerce platforms, or assess long-term implementation outcomes. Instead, the research concentrates on current stakeholder perceptions, lived experiences, and opportunities for improving digital tool usage on the shop floor. While this focus provides rich, context-specific insights, the findings may not be broadly generalizable to other manufacturing contexts or larger enterprises. These delimitations ensure the study remains focused and relevant to the specific challenges of SMEs undergoing digital transformation.

1.7 Structure of the Thesis

The thesis is structured as follows:

- **Chapter 1** introduces the background, problem formulation, research question, and methodology.
- **Chapter 2** reviews the literature on user experience (UX), service design, and digital transformation in manufacturing.
- **Chapter 3** presents the methodology, including case selection, data collection, and analytical framework.
- **Chapter 4** presents the empirical findings, organized into thematic categories developed through Reflexive Thematic Analysis.
- **Chapter 5** discusses the implications of the findings in relation to theory and practical design considerations.
- **Chapter 6** concludes the thesis with a summary of contributions, limitations, and suggestions for future research.

2

Literature Review

This chapter introduces the conceptual and empirical foundation relevant to the study of service-related UX factors and digital system adoption in traditional manufacturing environments. It covers core literature on digital transformation, user experience (UX) design, and service design, particularly within small- to medium-sized enterprises (SMEs). The review synthesizes insights from multiple disciplines, including manufacturing, healthcare, and construction, to identify critical gaps, emerging trends, and cross-sectoral strategies. It is organized into two main parts. The first part (Section 2.1) focuses on foundational theories and human-centered approaches that inform this study's design. The second part (Section 2.2) highlights highly relevant recent works that directly address key research challenges and align closely with the thesis objectives.

2.1 Theoretical Foundations and Human-Centered Approaches

This section outlines the theoretical basis for understanding digital system adoption through a service-oriented and human-centered lens. It brings together insights from service design theory, UX challenges in industrial contexts, and organizational dynamics that shape the adoption process. These foundations inform the methodological choices and empirical focus of this study.

2.1.1 Service Design as a Human-Centered Approach

Service design provides a structured yet flexible approach for aligning digital systems with user needs. Rooted in principles such as co-creation, holism, and iterative learning, it has traditionally been applied to customer-facing experiences [42]. However, its relevance for internal systems—particularly in manufacturing SMEs—is increasingly recognized.

Rather than treating internal tools as purely technical systems, service design conceptualizes them as part of a larger user journey involving workflows, expectations, and emotional responses. By mapping these journeys and involving users—such as machine operators and engineers—in the design process, organizations can identify frictions and co-develop more intuitive, acceptable solutions. This is especially critical in resource-constrained environments where training opportunities and tolerance for inefficiencies are limited.

Service design also facilitates cross-role understanding by visualizing both "frontstage" interactions (e.g., operator interfaces) and "backstage" infrastructure (e.g., database or process logic). Through shared tools such as service blueprints, stakeholders can uncover hidden misalignments and foster collaborative problem-solving. These characteristics make service design not only a usability methodology but also an organizational change enabler, positioning it as a core lens for this study.

2.1.2 Digital Transformation and UX Challenges in Manufacturing

Digital transformation in manufacturing is often associated with Industry 4.0 technologies, such as real-time dashboards, touchscreen systems, and barcode-based automation [17]. In theory, these tools promise enhanced traceability and streamlined operations. Yet, in practice, their implementation frequently encounters UX-related obstacles—especially in SMEs, where workflows tend to be informal and personnel resources limited.

Operators often experience cognitive overload from poorly designed interfaces, unclear system logic, or insufficient training [18]. When digital tools disrupt established routines without offering perceivable benefits, resistance increases [33]. Touchscreen-based workstations, for example, may delay production if navigation is too complex, prompting users to develop manual workarounds. These issues reduce not only system efficiency but also trust in digitalization as a whole.

Such cases underscore the importance of integrating UX thinking early in digital system design. In manufacturing environments, usability directly influences adoption and operational continuity, making it a strategic—not just technical—concern.

2.1.3 Human and Organizational Dynamics in Adoption

Adoption outcomes are shaped not just by interface design but also by the broader organizational environment in which systems are introduced. In SMEs, hierarchical distance between managers and shop-floor staff can exacerbate misalignments. Operators often lack channels to voice concerns or influence system adjustments, while managers may interpret low uptake as a training issue rather than a usability failure [45].

This disconnect leads to frustration, informal resistance, and long-term disengagement. Studies have shown that meaningful support structures—such as real-time assistance, contextualized onboarding, and participatory testing—can significantly improve adoption rates [10]. Yet, these are often missing in lean organizations.

Service design offers practical tools to address these gaps. Techniques such as blueprinting and role-based journey mapping encourage mutual understanding across functional boundaries. By visualizing the end-to-end experience and surfacing user pain points, these methods help synchronize strategic goals with day-to-day realities, facilitating smoother transitions.

These conceptual foundations establish the rationale for this study's methodological choices and analytical focus. The following section reviews empirical work that illustrates how these principles and challenges materialize across different organizational and industrial contexts.

2.2 Highly Relevant Work

This section reviews empirical studies that demonstrate how the theoretical gaps and challenges identified in Section 2.1 manifest in real-world settings. Drawing from diverse industries such as manufacturing, healthcare, and construction, these examples illustrate how service design, human-centered UX approaches, and cultural alignment influence digital system adoption. Each subsection builds on the previous discussion by showing how these principles translate into concrete practice.

2.2.1 Service Design and Human-Centered Approaches in Digital Transformation

As discussed in Section 2.1.1, service design offers a framework for aligning digital tools with user needs through participatory and holistic methods. Empirical studies provide growing evidence of its effectiveness in internal transformation efforts. One example examined ten European manufacturing firms transitioning toward servitization, where iterative design sessions and cross-functional workshops enabled the co-development of service-aligned digital tools [22]. The process revealed how early integration of service design facilitated consensus among stakeholders and minimized downstream resistance.

A complementary study applied service blueprinting to uncover friction points in internal workflows, leading to actionable system redesigns and improved operator alignment [4]. These findings reinforce the practical viability of service design not only in external service delivery, but also in restructuring internal processes. The present thesis builds on this foundation by applying journey mapping and co-creation workshops with frontline operators and managers in a traditional SME context.

2.2.2 UX Challenges and Adoption on the Shop Floor

Reflecting the UX barriers outlined in Section 2.1.2, several studies emphasize the critical impact of usability and contextual fit. In a study of fifteen Finnish manufacturers, many firms adopted advanced technologies without accounting for shop-floor realities, resulting in low engagement and the proliferation of informal workarounds [36]. These observations parallel this thesis's field findings, where operators sidestepped inefficient systems through manual data recording and peer-dependent troubleshooting.

Another case from an Italian assembly plant involved operators in designing the interface of a collaborative robot and AGV system. Their involvement led to a more intuitive system, reducing training time and increasing adoption [34]. This example underscores how participatory UX methods can lead to tangible usability gains in industrial settings.

Parallel issues have been documented outside manufacturing. In a multi-site study of Swiss hospitals, medical staff experienced mismatches between new digital systems and their workflow expectations. Usability issues such as information overload and fragmented interfaces emerged, highlighting that internal UX challenges transcend sectors [46]. These findings further validate the thesis's application of a human-centered approach to internal digital systems in SMEs.

2.2.3 Cross-Role Communication and Cultural Factors in Adoption

Expanding on the organizational misalignments discussed in Section 2.1.3, cross-role communication and organizational culture emerge as decisive factors in adoption success. A study of Irish MedTech companies revealed that firms promoting a human-centric culture—with open communication and shared learning—achieved deeper transformation outcomes than those focusing purely on technical rollout [37]. These results support the thesis's emphasis on cultivating cross-level dialogue to accompany UX and service design efforts.

Similarly, a study in healthcare found that excluding clinicians from the system selection process resulted in low trust and partial adoption of new tools [46]. This mirrors the top-down digitalization challenges faced by manufacturing SMEs, where shop-floor voices often go unheard.

Conversely, in a construction industry case, engaging site workers in co-specifying digital tools led to a smoother rollout and higher satisfaction [32]. These findings affirm that participatory strategies not only enhance UX but also foster organizational alignment, which is especially valuable in SMEs with flat hierarchies and resource constraints.

3

Research Design and Philosophical Approach

This chapter presents the methodological framework that shaped the study's inquiry into how service-related factors influence UX and digital system adoption in a traditional manufacturing SME. It details the epistemological orientation, case context, multi-method data collection, and interpretive strategies used to explore the lived realities of operators, engineers, and managers during digital transformation.

3.1 Research Design and Theoretical Framework

The study employs a qualitative, embedded single-case study design to investigate the interplay of service-related factors, UX, and digital system adoption in a manufacturing SME. This design enables deep engagement with stakeholder experiences within a specific organizational setting [47]. Rather than seeking generalizable truths, it aims to uncover situated insights and contextual dynamics that shape digital adoption in real-world environments.

3.2 Qualitative Research Rationale

A qualitative orientation is well-suited to exploring socially embedded phenomena such as digital transformation, where human experiences, perceptions, and organizational structures co-evolve [15]. This approach enables the researcher to probe both observable interactions (e.g., touchscreen use) and participants' underlying meanings and emotions—such as anxiety over irreversible inputs or satisfaction from self-reliant problem-solving. It supports the thesis's focus on experience, emotion, and service quality as central dimensions of digital adoption [7].

3.3 Case Study Approach

This research adopts an embedded single-case study strategy, selecting a medium-sized wood furniture factory as a bounded system [41]. The embedded design allows for the exploration of multiple stakeholder perspectives—operators, engineers, and managers—within the same organizational setting. Case study methodology excels in integrating multiple data sources to capture complex socio-technical dynamics,

making it particularly suitable for examining the intersections of service design, digital tools, and shop-floor realities [47].

3.4 Philosophical Paradigm: Interpretivism

The study is grounded in an interpretivist paradigm, which assumes that knowledge is constructed through human interpretation and shaped by context [39]. Interpretivism aligns with service design principles, which emphasize empathy, co-creation, and the subjective quality of user experience [42]. Rather than measuring system effectiveness in abstract terms, the study prioritizes understanding how stakeholders perceive, feel, and respond to digital interventions.

The interpretivist stance also acknowledges the researcher's influence in shaping meaning. Working in a Chinese SME setting as a bilingual researcher trained in European design traditions, I approached the field as both an insider and an outsider. This perspective allowed me to access nuanced insights while remaining critically reflective about my interpretations.

3.5 Reflexivity and Positionality

As the sole researcher, my identity shaped every stage of the inquiry—from field access to analysis. I was fluent in Mandarin and socially attuned to the local setting, which facilitated rapport-building and natural communication. At the same time, my methodological training in Western service design approaches enabled a critical lens that occasionally diverged from managerial or engineering perspectives within the factory.

All interviews and workshops were conducted in Mandarin and later translated by me into English. Translation required close attention to tone, context, and culturally embedded meanings. Throughout this process, I maintained a reflexive journal to document key moments—such as when participants hesitated to speak in groups, or when terminology needed clarification. This ongoing reflexivity helped surface potential biases and enriched the interpretive depth of the analysis.

Rather than striving for detached neutrality, I approached my role as a co-creator of meaning. My interpretations were shaped through direct engagement and cultural navigation, grounded in empathy as well as analytical awareness.

3.6 Case Context and Stakeholder Landscape

The study was conducted in a medium-sized wood furniture factory in Anhui, China, employing approximately 180 workers across production stages (cutting, sanding, assembly, painting, packaging, quality control). As a typical SME, the factory operates with a lean management structure and cost-sensitive practices, reflecting the incremental digital transformation common in China's manufacturing sector [18].

3.6.1 Digital Transformation in the Factory

The factory is currently in the middle of a five-year digital transformation initiative aimed at improving traceability, minimizing paper-based processes, and increasing

workflow visibility across departments. As part of this initiative, several key systems have been introduced. Touchscreen interfaces were installed at workstations to enable operators to track job progress and enter production data in real time. Barcode scanners are now used for identifying materials and managing logistics across different production stages. In parallel, managers access a cloud-based dashboard system that aggregates live data for monitoring overall production flow and performance metrics.

These systems were developed through collaboration with an external vendor and further customized by the factory's internal engineering team. While technically functional, their implementation has largely favored managerial and system-level requirements over end-user usability. Operators were not involved in the design process, and no formal UX testing or participatory feedback loops were established prior to rollout. As a result, several mismatches have emerged between system functionality and everyday production practices. For example, operators often reported that slow-loading interfaces disrupted their workflow, while unclear feedback on data submission led to duplicate entries or missed inputs [3].

3.6.2 Organizational Structure and Stakeholder Roles

The factory operates through a hierarchical, functionally segmented structure. Three primary stakeholder groups shape and are shaped by the digital systems under investigation.

Operators, positioned at the production line, are responsible for real-time data entry via touchscreen interfaces and barcode scanners. Their tasks are standardized but time-sensitive, and they frequently encounter usability challenges—such as delays, unclear feedback, or system crashes—that directly impact their ability to meet quotas. Many operators expressed that their concerns often go unheard by management, leading them to develop informal workarounds or rely on peer knowledge-sharing.

Engineers act as system administrators and technical support personnel. Most are trained in IT or automation, and they prioritize stability, data integrity, and maintainability over front-end usability. They often bridge between the software vendor and the realities of the factory floor, though their proximity to day-to-day user struggles varies. Engineers are aware of the technical debt in the system but tend to view issues as solvable through training rather than redesign.

Managers occupy a more strategic layer. They monitor operations through aggregated dashboards and use system data for planning, procurement, and performance evaluation. Managers typically believe that digital tools increase traceability and efficiency, though they are often unaware of how these systems are actually used—or bypassed—on the shop floor. The lack of routine feedback mechanisms creates a gap between perceived and actual system performance.

These roles interact within a tightly interdependent but asymmetrical communication structure. Understanding their differing expectations and priorities was essential for interpreting both interview data and workshop outcomes.

To clarify the distribution of perspectives included in the interviews, Table 3.1 provides an overview of participant roles, experience levels, and relevant contextual information.

Table 3.1: Overview of Interview Participants

Pseudonym	Role	Experience (Years)	Gender	Relevant Background
Operator A	Operator	2	Male	Works on sanding line; often mis-scans and resorts to paper notes
Operator B	Operator	5	Female	Assists peers with system use; skips unclear features
Operator C	Operator	3	Male	Experiences login errors; relies on repetition and help from others
Operator D	Operator	6	Male	Avoids touchscreen use during busy shifts; records manually
Operator E	Operator	4	Female	Prefers asking team leader rather than reporting via system
Operator F	Operator	4	Male	Hesitates to report issues; relies on peer confirmation due to lack of system response
Operator G	Operator	5	Male	Frustrated with engineering detachment; perceives system as disconnected from shop-floor reality
Engineer A	Engineer	6	Male	Maintains backend infrastructure; emphasizes data integrity
Engineer B	Engineer	3	Male	Develops cloud dashboards; rarely interacts with operators
Engineer C	Engineer	5	Male	Troubleshoots frontend issues; views training as key to adoption
Manager A	Manager	8	Female	Oversees production; sees system as vital for planning
Manager B	Manager	10	Male	Senior manager; tracks KPIs, unaware of usability challenges
Manager C	Manager	7	Male	Believes system issues stem from user resistance
Manager D	Manager	9	Female	Supports more training; sees improvement in data flow

3.6.3 Rationale for Case Selection

The case was selected based on three main criteria: representativeness, accessibility, and contextual richness.

First, the factory typifies many manufacturing SMEs undergoing incremental digital transformation without strong UX or participatory design strategies. As such, it provided a realistic setting to explore misalignments between systems and users. Second, my pre-existing relationship with a senior engineer at the factory—developed prior to this study—facilitated extensive access. This contact helped coordinate visits, secure permissions, and mediate introductions to participants. Third, the site's heterogeneity of roles and practices offered valuable variation. While some departments used the digital system rigorously, others relied on paper or hybrid methods. These contrasts enabled comparative insight into adoption challenges.

Furthermore, the shared language (Mandarin) allowed for open dialogue with most participants. There was no perceived risk of conflict of interest, as I had no prior involvement in the factory's operations. Instead, my position as an "informed outsider" enabled both access and analytical distance.

3.7 Data Collection Procedures

Data were gathered over a three-month period through a multi-method strategy designed to elicit diverse stakeholder experiences. Four qualitative methods were employed: semi-structured interviews, non-participant observations, a co-creation workshop, and service mapping.

Sixteen semi-structured interviews explored users' daily system interactions, onboarding experiences, usability frustrations, and cross-role communication. Interviews were conducted in Mandarin, recorded with consent, and translated by the researcher with attention to meaning retention and contextual nuance.

Field observations took place across five production stages during multiple shifts, capturing spontaneous interactions and real-time system use. Notes documented behavior patterns, informal adaptations, and social dynamics—such as reliance on peers or use of handwritten notes in place of system input.

The co-creation workshop brought together two participants from each stakeholder group to annotate journey maps and propose workflow improvements. This collaborative activity illuminated misalignments and fostered cross-role dialogue that seldom occurred in daily operations.

Finally, service blueprinting synthesized insights from the three methods above. This visual tool mapped out touchpoints, backstage processes, and user frustrations, supporting both thematic analysis and practical design implications.

3.8 Data Analysis and Trustworthiness

Data were analyzed using Reflexive Thematic Analysis (RTA), following the six-phase process [6, 7]. After familiarization and initial coding, patterns were iteratively reviewed and clustered into overarching themes.

Codes were generated inductively and refined through constant comparison across sources and stakeholder groups. For example, operators' descriptions of workarounds informed a theme on "invisible adjustments," while engineers' attribution of issues to user error contributed to "misaligned accountability."

Triangulation emerged naturally by comparing findings across methods. Interview insights were validated through field observations and echoed during the workshop. A research journal tracked analytic decisions, theme evolution, and researcher reflections.

Rather than applying generic validity checklists, the study emphasizes transparency, coherence, and interpretive depth. Translated quotes were member-checked for accuracy, and rich contextual detail was provided to support transferability.

3.9 Ethical Considerations and Data Governance

All participants were informed of the study's purpose and consented voluntarily to take part. Written and verbal consent was obtained for participation and audio recording. No sensitive personal data were collected.

All sessions were conducted in Mandarin and anonymized during transcription. Pseudonyms were used to protect identities, and any identifying details were removed or altered. All research data were securely stored on an encrypted device, accessible only to the researcher.

Ethical protocols were guided by respect for participant autonomy, confidentiality, and well-being. Engagement was designed to minimize power imbalances and ensure participants felt comfortable contributing openly.

4

Findings

4.1 Overview of the Analytical Themes

This chapter presents the findings from a qualitative single-case study exploring how service-related factors influence user experience (UX) and the adoption of digital systems in a traditional wood furniture factory in China. Drawing on semi-structured interviews, field observations, service mapping, and a co-creation workshop, the analysis followed the Reflexive Thematic Analysis (RTA) approach [6, 7, 8]. The aim was to identify recurring tensions and patterns in how users encounter, adapt to, and work around digital technologies in their daily routines.

The findings are organized into four interrelated themes, each representing a key service-related factor that shaped both how the digital system was experienced and how it was adopted (or resisted) by different stakeholders. These themes were developed through an iterative process of inductive coding and synthesis across multiple data sources. They are presented in a sequence that reflects how foundational structural problems cascade into usability challenges, emotional strain, and broader organizational disconnects.

The first theme, *system–workflow misalignment*, examines how digital tools often failed to align with the actual rhythms and constraints of shop-floor work. The second theme, *feedback gaps and error anxiety*, explores how the lack of system responsiveness created hesitation, rework, and mistrust. The third theme, *learning curves and cognitive load*, highlights the mental demands of navigating unfamiliar digital systems without adequate support. Finally, the fourth theme, *cross-role misunderstandings*, reveals how divergent interpretations of system purpose and logic among operators, engineers, and managers contributed to miscommunication and stalled improvements.

Table 4.1: Overview of Analytical Themes

Theme	Description
System-Workflow Misalignment	Digital systems are not aligned with actual production sequences, leading to skipped inputs, retroactive entries, and informal workarounds that compromise data integrity.
Feedback Gaps and Error Anxiety	The lack of timely and clear system responses results in uncertainty, repeated inputs, and fear of making unfixable errors, causing stress and disengagement.
Learning Curves and Cognitive Load	Limited onboarding, minimal error guidance, and unfamiliar interface conventions increase the mental effort required to operate digital tools.
Cross-Role Misunderstandings	Stakeholders maintain siloed perspectives on system use, with engineers and managers often unaware of daily user challenges, limiting responsive improvements.

Table 4.1 summarizes these four themes and their central concerns.Each of the following sections (4.2–4.5) explores one of these themes in greater depth, supported by participant quotes, field observations, and visual materials from journey mapping and service blueprinting activities. Rather than isolated issues, these themes form a connected narrative that reveals deeper tensions in the digital transformation process—tensions that reflect service-related breakdowns across user support, workflow alignment, cognitive demand, and inter-role communication.

4.2 Theme 1: System–Workflow Misalignment

One critical service-related factor influencing user experience and digital system adoption was the persistent mismatch between the digital system's internal logic and the real-world workflows of frontline operators. While the system was intended to enhance traceability through standardized, linear steps, it often failed to accommodate the fluid, time-sensitive, and situationally adaptive nature of shop-floor work. This misalignment emerged clearly across interviews, field observations, and co-creation workshop sessions.

During an interview, an operator explained, "The scanner slows us down. We usually run the machine first and input everything later when it's quieter." This sentiment was further confirmed during direct observation in the sanding area, where the operator bypassed system prompts and used handwritten notes instead. These workarounds were not isolated incidents—participants frequently described how rigid digital sequences (e.g., scan before execution) conflicted with their practical need to prioritize physical production flow.

In the co-creation workshop, stakeholders from all three roles annotated a shared service blueprint of their production process. Engineers initially assumed real-time data entry was standard, only to realize that many inputs were deferred until the end of a shift. As one engineer remarked in the debrief, "We designed the sequence based on ideal steps. But the reality is more fluid. There was no feedback loop to tell us what actually happens."

To visualize these tensions, Figure 4.1 presents a service blueprint that maps each production phase to its respective goals, operator actions, system responses, and pain points.

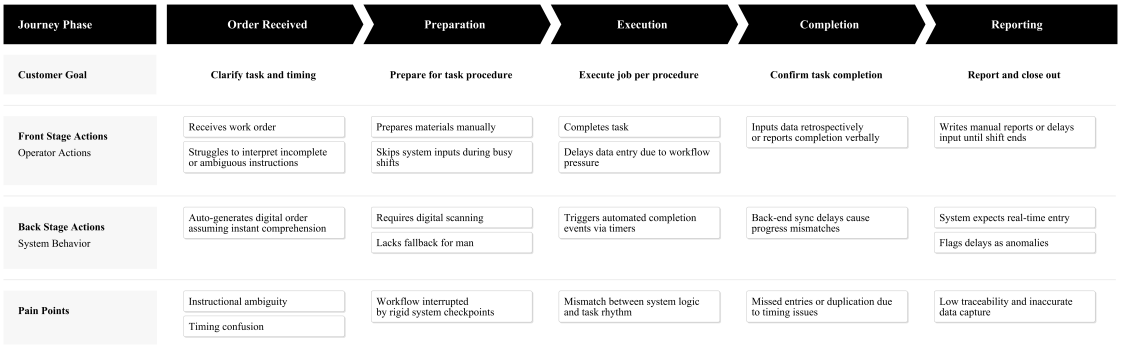


Figure 4.1: Service Blueprint: Misalignment Between Workflow and Digital System

This blueprint reveals not only technical design gaps but also service experience breakdowns—moments where the system failed to support user needs, prompting informal adjustments and disjointed data practices. What may appear as "non-compliance" from a managerial viewpoint is often a rational response to misaligned system demands.

These are not isolated usability bugs, but symptoms of a deeper orchestration failure between technological expectations and embodied human work. Addressing this challenge requires more than technical fixes or operator training; it demands participatory service design processes that continuously realign backend logic with actual workflows and temporal constraints.

These dynamics will be revisited in later themes—particularly in relation to feedback gaps and inter-role misunderstandings—and further interpreted through the lens of service blueprinting and organizational UX in Chapter 5.

4.3 Theme 2: Feedback Gaps and Error Anxiety

A second major theme that emerged from the data concerns the lack of immediate and clear feedback during digital system interactions. Both operators and engineers consistently reported uncertainty about whether their inputs—such as barcode scans or job completions—had been successfully registered or saved. This ambiguity often led to hesitation, redundant actions, and increased reliance on verbal confirmation or manual records. The issue was particularly acute in high-pressure production environments, where delays or doubts could disrupt workflow momentum.

During interviews, many operators shared their frustration with the system's silent failure modes. As one put it, *"Sometimes I press submit but the screen stays the same. I don't know if it worked, so I try again."* (Operator D). Another added, *"The system doesn't show if you scanned something wrong. It just doesn't respond. You think it's broken."* (Operator B). These experiences highlight a critical design flaw: the system lacked responsive micro-interactions that confirm user actions in real time. Operators were left guessing whether the machine had received their input, often resulting in duplicate entries or skipped steps.

This uncertainty was compounded by a lack of recovery options. Interviewees described how small mistakes—such as scanning the wrong barcode or missing a confirmation—could not be easily undone. *"If I make a mistake, I don't know how to fix it. I'm afraid it will mess up the records, so I call the engineer instead."* (Operator A). Others echoed similar concerns about the system's opacity: *"There is*

no history view or confirmation. You just guess what the last step was." (Operator E). In response to these limitations, many operators developed a cautious approach, avoiding certain system functions unless absolutely necessary and reverting to manual documentation when uncertain.

This lack of feedback and error recovery mechanisms not only interrupted operational flow but also affected operators emotionally. Several reported feelings of stress, anxiety, and lowered confidence when using the system. For example, one participant admitted that they sometimes avoided using the system altogether during busy periods to prevent mistakes. These sentiments reveal a broader problem: the system design placed an undue cognitive burden on users without providing adequate support for correction or reassurance.

From a service design perspective, these findings point to the absence of critical interactional affordances—such as confirmation cues, error visibility, and safe reversal pathways. Without these elements, digital tools become a source of friction rather than facilitation. A well-functioning internal system should not require users to second-guess their actions or fear irreversible errors.

The interviews, observations, and service mapping activities consistently revealed that the lack of feedback loops was a source of inefficiency and disengagement. Operators delayed data entry until the end of their shifts, engineers received incomplete or contradictory records, and managers were often unaware that data was entered retroactively. These gaps undermined the system's intended purpose of traceability and real-time monitoring.

Overall, feedback gaps and error anxiety did more than slow down individual tasks—they eroded trust in the system itself. Addressing these issues requires more than just interface tweaks; it demands a holistic reconsideration of how feedback is embedded into the user experience. Future design iterations should focus on improving the visibility of system status, integrating micro-confirmations for all actions, and offering intuitive pathways for users to detect and correct mistakes without expert intervention.

4.4 Theme 3: Learning Curves and Cognitive Load

Another prominent theme that emerged from the data was the high cognitive effort required to operate the digital system, especially for operators unfamiliar with similar technologies. Although many had years of experience with physical production tasks, they often struggled when switching to interface-based workflows. This friction was less about resistance and more about mental load—navigating system structures, remembering procedures, and managing uncertainty when errors occurred.

During an observation session in the sanding area, an operator repeatedly scanned a job card, but the system did not respond. The operator quietly wrote the job number on paper and continued working. When approached later, they explained that the screen "sometimes just freezes," and they "don't want to ask every time." This subtle disengagement reflects how usability issues are not always reported explicitly—they manifest as quiet workarounds.

Several interview participants described minimal or inconsistent onboarding experiences. Operator A explained: *"When I joined, a teammate showed me once. After that, I just guessed. If I'm not sure, I ask someone or skip it."* The lack of

formal training meant that many operators relied heavily on peer instruction or trial and error, especially when facing novel error states.

Even more experienced users reported confusion around system logic. In a post-interview debrief, Operator C explained: *"It's not that I don't want to use it. But sometimes I forget which screen or button to press, and then I worry I'll break something."* This anxiety about doing something irreversible often led users to either avoid certain functions or delay interaction until someone else could verify their steps.

These challenges were compounded by abstract interfaces that lacked visual cues or confirmation messages. As one engineer admitted during the co-creation workshop, "we expected users to just follow the steps. But the reality is more fluid." The mental gap between operator logic and system design was not bridged by onboarding or interface support. Instead, operators memorized sequences, made handwritten backups, or deferred interaction altogether.

Operator B, a five-year veteran of the assembly line, informally helped train newer colleagues but admitted: *"I usually help others, but even I don't know everything. When the screen lags or shows a strange message, I just skip it."* This reveals how even confident users lacked systemic understanding, and support was based on improvisation rather than structured learning.

System design did not account for differences in digital literacy or learning preferences. In the co-creation workshop, operators engaged more actively with visual journey maps than during verbal explanations. This suggests that current digital tools may not align with how operators process information—favoring visual, concrete representations over abstract menu logic.

STAGES	LOGIN & SETUP ▸	PREPARATION ▸	TASK EXECUTION ▸	DATA RECORDING ▸	TASK COMPLETION
EMOTIONS	Anxious	Confused	Frustrated	Hesitant	Relieved
GOALS	Start the task and access system	Prepare and verify materials	Execute task correctly and stay on time	Submit job data without making mistakes	Confirm output and wrap up task
EXPERIENCE	1. No onboarding 2. System UI unclear 3. Needed peer help	1. System skipped if scan forgotten 2. No prompt shown	1. Interface does not confirm execution 2. User unsure whether steps were saved	1. No feedback shown 2. Some users guess or defer to manual record	1. Final confirmation unclear 2. Reported completion verbally
THOUGHTS	"I don't want to keep asking."	"Did I do it right?"	"Did it register what I did?"	"Better write it down just in case."	"Hope they know I finished."
PAIN POINTS	1. Barcode scanner unclear 2. Login freeze 3. No visual cues	1. No prompt for skipped steps 2. Peer advice contradicts expectations	1. Submission freezes 2. No error messages	1. No history log or confirmation 2. Fear of making irreversible mistakes	1. Dashboard unclear 2. Completion logic inconsistent
COGNITIVE LOAD	1. Learn login steps from memory or peer 2. No structured training	Must switch between manual handling and system logic	Memorize multistep flow and interface logic	1. Must guess what has/ hasn't been submitted 2. Rely on peer review	No feedback loop to confirm submission success
OPPORTUNITIES	Embedded onboarding overlay with live guidance	1. Contextual prompt when scan is skipped 2. Allow flagging incomplete steps	1. Visual confirmation after submission 2. Allow "undo"	Feedback message with history log or task status tracking	Dashboard indicator for task completion & traceability

Figure 4.2: Journey map: operator experience with digital system interaction.

Figure 4.2 illustrates a typical operator's journey across the full task cycle, highlighting emotional responses, usability breakdowns, and design opportunities. This visual mapping complements the interviews and observations, surfacing points of mental overload and missed support during digital interaction.

In summary, the difficulty operators faced was not only a question of system functionality, but also of cognitive design. High memory demands, unclear flows, and fear of mistakes created a mental burden that discouraged active participation. These findings point toward the need for interfaces that are learnable, forgiving, and aligned with the way users make sense of digital systems—especially in high-pressure manufacturing contexts.

4.5 Theme 4: Cross-Role Misunderstandings

While usability problems were common across all system touchpoints, a deeper systemic barrier emerged from misalignments in how different stakeholder roles—operators, engineers, and managers—understood the digital system and its intended purpose. These misunderstandings led to mismatched expectations, delayed improvements, and recurring breakdowns in communication. A study revealed that so-called "design breakdowns" often occur when different professional roles—such as designers and developers—form divergent interpretations of how an interactive system should function[30]. These discrepancies, even when all parties are working toward the same objectives, can result in significant implementation gaps. In the context of this study, similar breakdowns were observed among operators, engineers, and managers. Although they shared an overarching goal of improving production through digital tools, their differing assumptions and expectations reflected deeper structural and cultural barriers to digital transformation within the organization.

4.5.1 Operators: Task-Focused and Risk-Aware

Operators judged the system primarily based on how it affected their day-to-day work. Their feedback emphasized immediate concerns such as speed, usability under pressure, and the risk of making irreversible mistakes. Many operators were hesitant to report problems, citing ambiguous channels or prior experiences of being ignored.

"I don't usually report problems unless it's really bad. Last time I did, they said just keep using it." (Interview with Operator F)

They also expressed frustration about perceived distance between those designing and those using the system:

"The engineer doesn't see what we see. They sit in the office and design it like it's perfect." (Operator G)

This task-oriented mindset fostered workarounds and peer-based strategies but also limited visibility into broader system goals.

4.5.2 Engineers and Managers: System-Centered and Outcome-Focused

In contrast, engineers focused on system performance, error rates, and infrastructure stability. From their perspective, silence meant success. As one engineer put it in a follow-up interview:

"We rarely get feedback unless something breaks. If no one complains, we assume it works." (Engineer A)

Engineers also described a growing tendency to restrict features based on past misuse:

"We had to lock certain features because operators used them incorrectly and corrupted data." (Engineer B)

Meanwhile, managers viewed the system primarily as a means to gain traceability and optimize production efficiency. Some acknowledged interface issues, but typically attributed them to operator behavior or lack of training.

"Sometimes the issue is just that people aren't used to the system. With time, they will adapt." (Manager B)

"From a management point of view, the system is very useful. We can track progress and generate reports." (Manager C)

These differing priorities—task execution, technical correctness, and managerial oversight—produced a fragmented picture of system performance. The system "worked" for some, while others silently struggled or disengaged.

4.5.3 Workshop Insights: Surfacing the Disconnects

These role-based misunderstandings became particularly visible during the co-creation workshop. The activity was structured around a collaborative annotation of printed user journey maps based on prior interviews. Each small group included one operator, one engineer, and one manager. They were asked to mark pain points, assumptions, and improvement ideas from their own perspective.

During the exercise, engineers frequently assumed operators preferred automation and speed, while operators described a need for manual flexibility and control. For instance, one journey segment indicated that barcode scanning should trigger an automatic completion flag. The operator annotated, "We often skip this step because it slows us down." The engineer replied, "But that breaks the traceability logic—we thought it helped you!"

In another exchange, a manager expressed surprise at the extent of manual reentry:

"I didn't know the operators had so many extra steps. We only see the final numbers." (Engineer C, during workshop)

Operators, in turn, explained the logic behind delaying input:

"Now I understand why they ask for changes. It's not because they don't want to use the system—it's because it slows them down." (Manager D)

A study observed that digital transformation efforts often suffer from "interpretative gaps," where different stakeholders ascribe conflicting meanings to the same system elements[30]. In this case, engineers regarded automation as stable and efficient, whereas operators experienced it as fragile and inflexible. Managers assumed real-time data entry was feasible, without recognizing the conflict this posed for production rhythm. What might be perceived as resistance, then, was often a rational adaptation to a system whose logic failed to align with the realities of frontline work.

Importantly, this was the first time that many of these perspectives were openly shared. The visual tools—particularly the annotated journey maps—enabled participants to externalize assumptions and engage with others' perspectives. Several

noted that the activity helped them "finally understand what the others are dealing with."

This theme underscores the need to treat internal systems not only as technical artifacts, but as shared experiences mediated by communication, trust, and role-specific constraints. From a service design standpoint, designing for meaningful adoption requires intentional scaffolding for feedback, transparency, and ongoing dialogue.

Future digital transformations should institutionalize such participatory exchanges, not only during redesign, but as a regular mechanism for cross-role calibration and mutual learning.

4.6 Summary and Thematic Interrelations

This chapter outlined four interconnected themes that emerged from interviews, observations, service blueprinting, and co-creation workshops. Rather than isolated usability issues, the findings revealed deeper systemic breakdowns in how digital systems were understood, used, and experienced on the shop floor of a traditional manufacturing firm in the midst of digital transformation.

Table 4.2: Thematic summary: emerging UX patterns and stakeholder impacts

Theme	Impact on System Use	Stakeholders Affected
Learning Curve and Cognitive Load	Operators struggle with login, navigation, and data input, relying on peers and workarounds due to limited onboarding and high interface complexity.	Operators
System-Workflow Misalignment	Tasks are often performed out of sync with system expectations, leading to skipped entries, retroactive corrections, and loss of traceability.	Operators, Engineers
Feedback and Visibility Gaps	Absence of real-time feedback or progress indicators causes uncertainty, repeat errors, and reduced trust in the system.	Operators
Communication Breakdowns	Role misalignment and fragmented information flows create delays, blame-shifting, and isolated process adaptations.	Operators, Engineers, Managers

Table 4.2 summarizes the themes, indicating how each impacted system use and which stakeholder groups were primarily affected.

The first theme highlighted the mismatch between system logic and actual production workflows, where digital processes failed to reflect the rhythm and logic of daily operations. Closely tied to this was the absence of clear feedback and recovery mechanisms—creating uncertainty, hesitation, and a lack of trust in the system. Operators often faced steep learning curves without adequate onboarding or guidance, leading to high cognitive load and reliance on informal workarounds. Compounding these issues were misalignments across roles: managers, engineers, and operators often held different assumptions and expectations, which hindered coordination and adaptation.

These challenges were deeply interwoven. For example, unclear system feedback amplified learning difficulties, while misaligned workflows often went uncorrected due to fragmented communication between roles. Together, they shaped not only the user experience but also the long-term credibility and adoption of digital tools.

Addressing such breakdowns requires more than technical fixes—it calls for redesigning systems to better fit real work practices, improving communication across roles, and building in user support from the start. The next chapter connects these insights with broader research in service design, industrial UX, and digital transformation to explore how traditional factories can pursue more human-centered innovation.

5

Discussion

5.1 Theoretical Integration of Findings

This section synthesizes the study's findings with existing research to answer the central question: what service-related factors influence user experience (UX) and the adoption of digital systems in traditional manufacturing environments? Drawing on insights from human-computer interaction (HCI), industrial UX, service design, and digital transformation, the analysis identifies four interdependent categories of service-related factors—emotional responsiveness, workflow alignment, cross-role coordination, and trust infrastructure—that directly shape how digital systems are perceived, experienced, and adopted.

5.1.1 Emotional Responsiveness and Cognitive Support

Recent developments in HCI and UX research emphasize that user experience is not merely a matter of interface usability, but a lived, emotional engagement with technology [29]. In industrial settings, operators experience technology under pressure, navigating system demands alongside physical fatigue, production targets, and unpredictable shop-floor dynamics. A study of process-control operators revealed a "rich tapestry" of emotional responses, from satisfaction in solving problems to frustration under high-stakes failure conditions [24].

These findings align with our data: operators frequently described anxiety and hesitation when using digital systems that lacked intuitive design, supportive feedback, or clear onboarding. Literature on positive UX reinforces this link, showing how emotionally attuned design can enhance well-being and foster engagement—even in constrained, non-office contexts [25].

Taken together, these insights identify emotional responsiveness and cognitive support as essential service-related factors. When overlooked, users may feel overwhelmed or disrespected, leading to avoidance or superficial compliance. Designing for emotional clarity and in-the-moment guidance is therefore central to digital adoption in SME settings.

5.1.2 Workflow Alignment and Service Continuity

From a service design perspective, system interactions must be coherently supported by back-end logic and workflow integration. Service failure often arises when the

backstage fails to support frontstage interactions [20]. In manufacturing, this includes systems that disrupt production flow, demand duplicate inputs, or offer unclear confirmations.

Our findings confirm this pattern. Operators frequently encountered mismatches between what the system required and what production rhythms allowed. For instance, complex barcode scanning steps during high-volume shifts led to informal workarounds, undermining system integrity.

Workflow misalignment thus emerges as a critical service-related factor. Without tight coupling between system processes and real shop-floor tasks, digital tools are perceived as detached or obstructive. Tools like service blueprinting can reveal these fault lines, guiding redesign toward continuity and responsiveness.

5.1.3 Cross-Role Coordination and Shared Understanding

Digital transformation reshapes organizational roles and expectations. However, when new systems are interpreted differently across stakeholders—managers, engineers, and operators—misalignment becomes a barrier. Role-specific goals and mental models must be synchronized to ensure successful change [14].

In our case, engineers emphasized data quality and technical uptime, managers focused on performance dashboards, and operators often felt disempowered or unheard. These divergent perspectives led to data manipulation, workaround cultures, and distrust.

This cognitive and communicative gap constitutes an organizational service failure. Structured cross-role interaction—through participatory workshops or joint reviews—can align expectations and build shared mental models [43]. Therefore, internal alignment and inclusive dialogue are vital service-related factors that underpin sustainable system adoption.

Additionally, hierarchical workplace culture played a subtle yet significant role in communication breakdowns. Several participants exhibited hesitation or softened their language when expressing frustrations with digital tools, especially when speaking about issues that might implicate management decisions. This hesitance likely stemmed from power asymmetries within the organization, where questioning system design could be construed as challenging authority. Such dynamics impeded open feedback loops, diminishing the potential for collaborative problem-solving during digital implementation.

5.1.4 Trust Infrastructure and Perceived Safety

Trust emerges as the foundational element in successful technology adoption. Studies in manufacturing have established trust as a critical success factor for Industry 4.0 implementations [43]. Other research identifies distrust as a primary adoption barrier [28].

Our analysis differentiates between two dimensions of trust: trust in technology (perceived system reliability) and trust in implementation (perceived fairness, competence, and intent of those deploying the system). In the studied Chinese SME, implementation trust was especially decisive—users' willingness to engage hinged more on their confidence in management's motives and responsiveness than on the system itself.

This underscores the need to approach digital adoption as a trust-building service journey rather than a purely technical deployment. Transparency, fail-safes, and ongoing support function as trust-enabling service features. When trust infrastructure is weak or absent, even highly functional tools face rejection. When it is strong, minor flaws are tolerated as part of an evolving system.

In summary, this chapter identifies four categories of service-related factors—emotional responsiveness, workflow alignment, cross-role coordination, and trust infrastructure—as decisive in shaping both user experience and system adoption in traditional manufacturing environments.

5.2 Practical and Design Implications

Building on the theoretical insights above, this section proposes concrete strategies to address the specific challenges uncovered Chapter 4, with each implication drawn directly from the lived experiences of operators, engineers, and managers.

The first priority is to improve onboarding and in-context guidance so that new users are not left to guess their way through unfamiliar features. Several operators described relying on peers to figure out basic steps, often avoiding parts of the system altogether. To address this, digital interfaces should include just-in-time support—such as tooltips or step-by-step overlays—that guide users through key actions when they need it. These micro-interactions transform training from a one-time event into an ongoing, embedded experience.

Second, systems should be better aligned with real production workflows using participatory process mapping. Our findings showed that operators often delayed scans or skipped digital checkpoints when under time pressure. To address this, co-creation sessions should involve all stakeholder roles in reviewing workflows and adapting system logic—for example, by enabling batch input or allowing offline entries that sync later. These adjustments reduce the need for informal workarounds and improve data accuracy.

Third, feedback and error recovery mechanisms must be treated as essential system features. Operators frequently cited avoidance behaviors triggered by system silence or irreversible errors. Adding visual confirmations, undo functions, and contextual help can improve transparency and build user confidence, encouraging more complete and accurate system use.

Finally, communication across roles should be formalized and recurring. The co-creation workshop demonstrated how shared journey mapping helped align perspectives between managers, engineers, and operators. Organizations should build on this by implementing regular multi-role review sessions or cross-functional retrospectives that create ongoing feedback loops and shared understanding.

Taken together, these design and implementation strategies reframe digital transformation as a collaborative, service-centered process—one that values alignment between systems and everyday practices, and builds trust through participation and responsiveness.

5.3 Methodological Reflection

This research adopted a qualitative, interpretive, single-case design, grounded in service design methods and reflexive thematic analysis. The aim was not only to understand the technical use of digital systems but also to explore the emotional, social, and organizational dynamics embedded in everyday work. This approach enabled a situated, experience-driven view of how operators, engineers, and managers interacted with digital tools in a traditional manufacturing SME.

A major strength of the methodology lies in its contextual depth. Over three months of fieldwork, I was immersed in the production environment, allowing for first-hand observation of informal routines, unspoken adaptations, and tacit knowledge-sharing—insights that would be difficult to access through surveys or retrospective interviews alone. By triangulating data from semi-structured interviews, field observations, co-creation workshops, and service mapping, the study was able to identify and validate recurring patterns across stakeholder groups.

Centering the participants' perspectives also proved critical. Instead of evaluating system performance through technical benchmarks, the research foregrounded the lived experiences of frontline users. Operators' hesitations, emotional responses, and workarounds were treated as important indicators of system misalignment. At the same time, the interpretive lens helped surface how engineers and managers understood usability and success—often in ways that contrasted with shop-floor experiences.

That said, the methodology also involved limitations. Conducting a single-case study naturally limits generalizability. While the selected factory shares many traits with other SMEs navigating digital transformation, the findings may not directly apply to larger firms, different industries, or alternative cultural settings. Nonetheless, the detailed empirical descriptions and thematic depth offer valuable insights that can be transferred to comparable contexts.

Language and translation presented an additional layer of complexity. All interviews and workshop sessions were conducted in Mandarin, allowing participants to speak freely and comfortably. The recordings were later transcribed and translated into English by the researcher specifically for the purpose of writing the thesis. While this helped preserve linguistic nuance and enabled close engagement with the material, the translation process carried the risk of interpretive shifts—especially in capturing emotional tones or culturally specific expressions that may not translate precisely.

Finally, my positionality also influenced the research process. As both interviewer and translator, and as a researcher trained in European service design traditions but culturally embedded in the Chinese context, I navigated the field from a dual perspective. While shared language and cultural familiarity helped build trust and access, my academic stance inevitably shaped how meaning was co-constructed. To remain critically aware of this influence, I kept a reflexive journal throughout the fieldwork—documenting moments of clarity, misunderstanding, or bias, and supporting an ongoing dialogue between my role and the data.

5.4 Future Research Directions

Among the possible avenues for future work, one direction stands out as particularly critical and generative: understanding cross-role communication as a foundational infrastructure for digital system adoption. While prior research often attributes failed implementation to user resistance or insufficient training, this study reveals that misalignments between stakeholder roles—operators, engineers, and managers—are a deeper and more persistent source of friction. These misalignments reflect not only diverging priorities but also a lack of shared understanding about how digital systems are meant to support everyday work.

Our findings show that when communication is fragmented and feedback loops are absent, digital tools become symbols of distrust or disconnection. Operators circumvent formal input procedures; engineers dismiss usability concerns as user error; and managers remain unaware of the lived challenges on the shop floor. These dynamics cannot be solved through interface design alone. They require organizational structures that support inclusive dialogue, mutual translation of needs, and shared mental models.

Future research should therefore focus on how cross-role communication can be operationalized as a designable element of digital transformation. For example, longitudinal interventions could test structured dialogue formats—like recurring journey-mapping sessions or role-integrated service reviews—and measure their effects on system adoption, trust, and performance. Rather than treating communication as facilitation or afterthought, such work would reposition it as core infrastructure for co-created systems. Comparative studies across industries and cultures could also illuminate how different hierarchies, norms, and trust dynamics shape this process.

While other directions—such as reframing internal systems as emotionally supportive services—remain valuable, this study makes clear that without cross-role coordination, even the most thoughtfully designed tools will struggle to gain traction. Digital transformation in traditional manufacturing must therefore be understood not only as a technical or UX challenge, but as a communicative and organizational one. Investing in this layer promises to yield more durable, inclusive, and context-sensitive pathways to digital adoption.

6

Conclusion

This thesis set out to explore how service-related factors shape user experience and affect the adoption of digital systems in traditional manufacturing. Through a qualitative case study in a Chinese furniture factory, guided by service design and reflexive thematic analysis, the research uncovered four key tensions: steep learning curves, misaligned workflows, missing feedback, and breakdowns in cross-role understanding. These challenges show that digital transformation often introduces new frictions, rather than simply improving existing routines.

Conceptually, the study reframes industrial UX as more than an interface issue. Experience with digital tools is relational—shaped by trust, emotion, and how well systems align with the flow of daily work. Adoption depends not just on functionality, but on how tools fit into lived realities.

Practically, the findings point to key design priorities: co-creating with frontline workers, embedding clearer feedback and recovery features, and sustaining dialogue across roles. While the single-case approach offered deep insight, it was shaped by its local context, timeframe, and cross-linguistic interpretation.

What became most clear is that digital transformation isn't just about technology—it's about how people adapt, relate, and find meaning in change. When systems are designed with this in mind, factories can evolve not only in efficiency, but in humanity and resilience.

Reflecting on the process, my understanding of digital change shifted. I started focused on systems and workflows, but the most revealing moments came from informal conversations, unexpected resistance, and watching how people adapted tools to their needs. I came to see adoption as deeply social and emotional.

If I could begin again, I would spend more time understanding how each role—managers, engineers, and operators—sees the system differently. I'd also value casual moments more, from break-room chats to small talk on factory tours. These were not side notes; they were central to the story.

In future work, I hope to involve participants earlier—perhaps co-framing research questions or opening space for dialogue before formal methods begin. If internal systems are services, then research must be a service too: responsive, empathetic, and attuned to context. That mindset now shapes how I study—and how I want to show up as a researcher.

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Appendix A

(Interview Question Guid)

This appendix provides the complete set of semi-structured interview questions used in the field study. Questions were customized for each stakeholder group—operators, engineers, and managers—with the aim of eliciting insights on digital system use, usability challenges, learning processes, emotional responses, and cross-role communication in the context of digital transformation.

A.1 Questions for Frontline Operators

- Can you describe how you typically use the digital system during your shift?
- Are there moments when you choose to delay or skip using the system? Why?
- What parts of the system do you find most confusing or mentally exhausting?
- How were you trained to use the system? Was it helpful?
- What do you do when the system doesn't respond or gives you unclear feedback?
- Have you ever made a mistake while using the system? How did you fix it?
- Do you feel confident using all parts of the system? Why or why not?
- What kinds of support (from peers, supervisors, engineers) do you rely on?
- If you could change one thing about the system, what would it be?
- Do you report system issues? If not, what stops you from doing so?

A.2 Questions for System Engineers

- What are your responsibilities regarding the design and maintenance of the system?
- How do you typically receive or collect feedback from operators or managers?
- Are there features that frequently cause errors or complaints? Why do you think that is?
- How do you balance technical stability with ease of use for non-technical users?

- Do you think operators use the system as intended? If not, what are the causes?
- How do you interpret silent or non-reported issues? Is no news always good news?
- Have there been any cases where operators used the system in unintended ways?
- How is usability considered during design or updates? Are users involved in that process?
- In your opinion, what are the biggest obstacles to improving the system?

A.3 Questions for Managers

- What goals are you trying to achieve with the digital systems in production?
- How do you evaluate whether the systems are working well?
- How do you get information about system use on the shop floor?
- Do you feel the system is used consistently across all teams? Why or why not?
- When there are problems with usage, how do you determine the root cause?
- In your view, are challenges mainly about user training, design issues, or workflow fit?
- Have you ever adjusted management practices based on system feedback or reports?
- Do you talk directly with operators about their experiences using the system?
- What would you like to see improved in the way the digital system supports production?

These questions were used flexibly during interviews, allowing participants to elaborate freely and bring in their own experiences. All interviews were conducted in Mandarin and later translated into English for analysis.

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